

UNDERKHAAN, Mongolia

WMO: 443040

Lat: 47.304N

Lon: 110.608E

Elev: 3389

StdP: 12.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-28.2	-37.4	0.7	-32.0	-33.2	1.0	-27.9	26.6	2.8	22.8	3.7	6.8	230	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	89.8	62.0	85.7	60.4	82.1	59.6	65.6	80.0	63.9	77.7	62.5	75.9	12.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	92.1	69.2	59.6	86.5	68.2	57.8	81.0	67.7	32.4	80.3	31.1	77.6	29.9	75.8	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.5	26.0	21.7	-36.5	95.6	4.5	4.1	-39.8	98.5	-42.4	100.9	-45.0	103.2	-48.3	106.2		
(5)			-36.4	69.2	4.5	2.0	-39.6	70.6	-42.2	71.8	-44.7	72.9	-48.0	74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	-9.5	-0.5	18.8	38.9	52.0	63.7	68.9	64.5	52.5	35.0	11.4	-4.7
	DBStd	28.96	9.40	11.04	12.49	8.77	8.53	6.80	5.79	6.22	8.32	9.26	13.17	9.30
	HDD50	8038	1844	1414	966	346	78	1	0	0	67	468	1157	1697
	HDD65	12056	2309	1834	1431	782	409	104	23	84	381	930	1607	2162
	CDD50	1748	0	0	0	14	141	412	585	450	142	4	0	0
	CDD65	289	0	0	0	0	7	64	143	68	7	0	0	0
	CDH74	3704	0	0	0	9	255	946	1539	815	140	0	0	0
	CDH80	1275	0	0	0	2	69	332	581	268	23	0	0	0
Wind	WSAvg	8.8	8.0	8.3	9.1	11.4	11.8	9.3	7.7	7.1	8.4	8.4	7.9	8.0
Precipitation	PrecAvg	9.6	0.1	0.1	0.1	0.3	0.7	1.9	2.6	2.2	1.0	0.4	0.1	0.1
	PrecMax	16.2	0.4	0.5	0.6	1.1	1.6	6.5	11.3	7.6	3.4	1.5	0.5	1.3
	PrecMin	4.8	0.0	0.0	0.0	0.0	0.1	0.3	0.1	0.3	0.0	0.0	0.0	0.0
	PrecStd	2.6	0.1	0.1	0.1	0.3	0.4	1.2	1.9	1.6	0.7	0.4	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	35.9	57.6	73.5	87.5	93.3	96.5	92.7	83.2	68.2	46.6	24.4
		MCWB	17.7	28.3	39.3	48.0	55.7	60.1	64.0	62.4	56.4	47.6	35.5	20.7
	2%	DB	14.2	28.8	49.5	66.4	80.7	87.9	90.6	86.7	78.0	61.3	40.5	17.8
		MCWB	11.9	23.2	35.7	45.1	53.7	58.9	63.8	61.6	55.4	44.3	31.8	15.3
	5%	DB	9.5	22.8	43.4	61.5	75.8	83.6	86.0	82.2	73.5	55.7	35.1	13.8
		MCWB	7.6	18.9	32.3	42.9	51.5	58.2	62.4	60.5	53.6	41.6	28.2	11.7
	10%	DB	5.3	17.6	38.0	56.7	71.0	79.2	82.3	78.2	68.7	51.1	30.3	9.9
		MCWB	3.9	15.0	29.4	40.4	49.5	57.0	61.8	59.3	51.8	39.0	25.3	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	28.5	40.0	49.2	57.5	64.5	69.1	67.1	61.0	48.3	35.7	21.0
		MCDB	20.8	35.7	56.6	68.6	80.0	81.5	85.4	80.9	73.8	65.0	45.9	24.6
	2%	WB	11.9	23.3	35.9	46.0	54.6	62.2	66.5	64.7	58.3	45.1	32.0	15.4
		MCDB	14.1	28.5	48.2	64.2	77.2	79.3	82.2	77.2	70.2	59.4	40.0	17.6
	5%	WB	7.6	19.1	32.5	43.2	52.4	60.4	64.9	63.1	56.0	42.1	28.6	11.8
		MCDB	9.4	22.8	42.7	60.2	72.9	76.4	79.5	75.7	68.4	54.7	34.5	13.7
	10%	WB	3.9	14.9	29.4	40.5	50.1	58.9	63.5	61.5	53.7	39.4	25.3	8.3
		MCDB	5.3	17.7	37.4	55.5	68.3	74.8	77.2	73.8	66.7	50.4	30.2	9.9
Mean Daily Temperature Range	5% DB	MDBR	22.3	27.4	27.1	26.8	27.9	24.4	22.1	23.2	25.8	24.9	23.1	21.0
		MCDBR	25.1	30.8	32.5	35.0	36.8	31.8	28.9	30.2	32.3	32.2	26.5	23.4
		MCWBR	23.1	26.0	21.0	19.3	16.7	12.4	9.5	11.8	14.9	19.0	19.3	21.1
	5% WB	MCDBR	25.0	30.6	31.1	32.6	33.0	25.6	23.4	23.6	26.3	30.3	26.3	23.2
		MCWBR	23.0	26.0	20.6	18.6	15.9	11.0	9.3	10.5	12.8	18.2	19.3	20.9
Clear-Sky Solar Irradiance	taub		0.247	0.271	0.309	0.364	0.416	0.431	0.396	0.371	0.331	0.293	0.262	0.247
	taud		2.338	2.299	2.270	2.173	2.060	2.130	2.302	2.342	2.389	2.427	2.390	2.326
	Ebn at Noon		272	286	289	281	269	265	273	276	279	275	263	256
	Edh at Noon		22	29	36	43	50	48	40	36	31	26	21	20
All-Sky Solar Radiation	RadAvg		622	983	1468	1849	2102	2070	1985	1809	1479	1045	670	497
	RadStd		42	63	56	67	95	128	89	92	86	60	36	38

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page